



Anatomy-inspired model for critical landmark localization in 3D spinal ultrasound volume data

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ABSTRACT

Three-dimensional (3D) spinal ultrasound imaging has demonstrated its promising potential in measuring spinal deformity through recent studies, and it is more suitable for massive early screening and longitudinal follow-up of adolescent idiopathic scoliosis (AIS) compared with X-ray imaging due to its radiation-free superiority. Moreover, some deformities with low observability, such as vertebral rotation, in X-ray images can also be reflected by critical landmarks in 3D ultrasound data. In this paper, we propose a localization network (LLNet) to extract lamina in 3D ultrasound data, which has been indicated as a meaningful anatomy for measuring vertebral rotation by clinical studies. First, the LLNet skillfully establishes a parallel anatomical prior embedding branch that implicitly explores the anatomical correlation between the lamina and another anatomy with more stable observability (spinous process) during the training phase and then introduces the correlation to highlight the potential region of the lamina in the inferring one. Second, since the lamina is a tiny target, the information loss caused by continuous convolutional and pooling operations has a profound negative effect on detecting the lamina. We employ an optimization mechanism to mitigate this problem, which refines feature maps according to information from the original image and further reuses them to polish output. Furthermore, a modified global-local attention module is deployed on skip connections to mine global dependencies and contextual information to construct an effective image pattern. Extensive comparisons and ablation studies are performed on actual clinical data. Results indicate that the capability of our model is better than other outstanding detection models, and functional modules effectively contribute to this, with a 100.0 % detection success rate and an 8.9 % improvement of mean intersection over the union. Thus, our model is promising to become a helpful part of a computer-assisted diagnosis system based on 3D spinal ultrasound imaging.

1. Introduction

Adolescent idiopathic scoliosis (AIS) is likely to progress rapidly without any interventions, and severe AIS can seriously affect a patient's daily life. In this case, early screening and frequent monitoring of adolescents play a meaningful role in timely diagnosis and treatment. The Cobb angle measurement in standing coronal X-ray images is the gold standard for evaluating spinal deformity in most clinical practices (Cobb, 1948). Unfortunately, X-ray imaging is unsuitable for frequently monitoring AIS patients due to the radiation exposure (Levy et al., 1996). Furthermore, scoliosis is usually a three-dimensional (3D)

deformation consisting of lateral curving and vertebral rotation, but the vertebral rotation cannot be observed in coronal view effectively.

Recently, it is heartening to note that radiation-free 3D ultrasound imaging has been successfully utilized in visualizing spinal deformity (Cheung et al., 2013; C Cheung et al., 2015). As demonstrated in Fig. 1, a sequence of cross-sectional slices on transverse plane consists of 3D spinal ultrasound volume data, and a typical slice is displayed in Fig. 1. The bone surface with a high acoustic impedance significantly reflects ultrasound waves, so a typical “W” contour exists in the slice due to the vertebral anatomy as explained in Fig. 1. Through Volume Projection Imaging (VPI) for a complete 3D spinal ultrasound volume data (C

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Cheung et al., 2015), a coronal VPI image visualizes some critical anatomical structures, and the spinal profile can be observed effectively corresponding to coronal X-ray images (Huang et al., 2022; Banerjee et al., 2024). Then, like Cobb angle measurement, a counterpart called ultrasound curve angle (UCA) in coronal VPI image is conducted to quantify the degree of spinal deformity without radiation, which is a representative application for 3D spinal ultrasound imaging. Many studies have also indicated a high correlation between the Cobb angle and UCA (Wang et al., 2019; Pang et al., 2021). In addition to this, the 3D advantage of spinal ultrasound imaging provides more possibilities for a comprehensive assessment of the spine than x-ray images with 2D plane transmission imaging. Similarly, the 3D volume data can be projected on the sagittal view to visualize the spinal profile, and it has been reported that a correlation exists between the UCA and Cobb angle measurement on sagittal VPI and X-ray images (Lee et al., 2021).

Unlike the capability of coronal and sagittal views for assessing the spinal profile, the transverse view has a unique potential to measure vertebral rotation thanks to detected laminae (Lee et al., 2023). Before 3D spinal ultrasound imaging emerged and was employed in medical practices, the EOS® system achieved a promising vertebral rotation evaluation through spinal 3D reconstruction based on biplanar 2D radiographs. Nevertheless, the reconstruction is time-consuming, and the costly expenditure and radiation hazards are considerable. Computed tomography (CT) and magnetic resonance imaging (MRI) also face similar problems, and the two devices work on non-weight-bearing positions, namely, subjects are usually scanned lying down, which is not conducive to assessing the case of the spine (Birchall et al., 1997; Hasegawa et al., 2018). By contrast, 3D spinal ultrasound imaging can outperform vertebral rotation assessment on the transverse view. As shown in Fig. 1, the lamina is also a typical component on the “W” contour. Promisingly, manual annotation of lamina on spinal ultrasound transverse slices achieves a practical vertebral rotation evaluation, and a high correlation exists between the measured result based on the 3D spinal ultrasound imaging and that of the EOS system (Lee et al., 2023). However, a complete 3D spinal ultrasound volume data contains numerous transverse slices. Thus, the manual annotation of lamina is labor-intensive and impractical in extensive medical practices. A fast and fully automatic localization approach for lamina is required to evaluate the vertebral rotation based on 3D spinal ultrasound highly efficiently, but there are still some technical challenges. First, the size of the region of interest (ROI), including the lamina, is defined as 20 pixels $\times$ 20 pixels, depending on advices from clinical experts and technical staffs. By contrast, the size of an entire ultrasound transverse slice is 256 pixels $\times$ 256 pixels, so this task is a localization for a tiny target. Second, some cases, such as inadequate contact between the probe and the subject’s back, negatively affect image quality, as highlighted in Fig. 2.

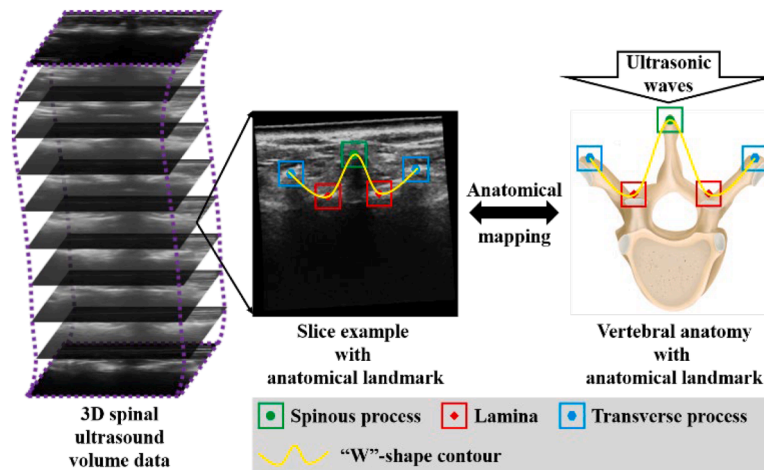


Fig. 1. Example of a cross-sectional slice from 3D spinal ultrasound volume data and its anatomical map.

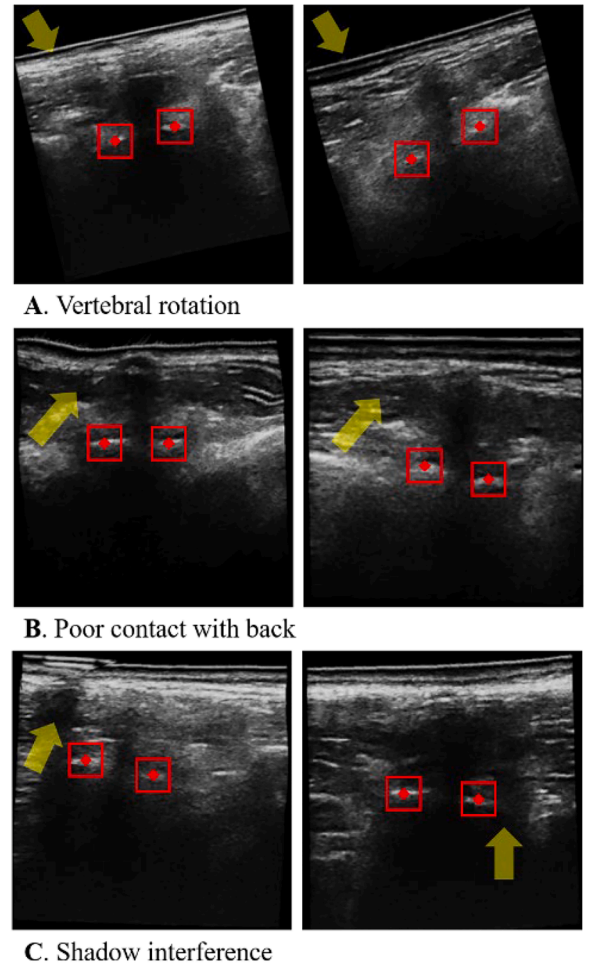


Fig. 2. Examples of typical interferences existing in transverse slices from 3D spinal ultrasound volume data. Red bounding box and point indicate the position of lamina. Yellow arrow highlights the interference.

Third, due to the large number of slices contained in a 3D spinal ultrasound volume data, the time cost of laminal detection should be considered to suit the massive early screening and longitudinal follow-up.

Traditional detection or tracking algorithms, like those relying on the optical flow field and local similarity, usually require manual

initialization and are difficult to adapt to image diversity and variability. Fortunately, deep learning approaches are driven by data, so they perform robustly in complex and changing situations, and many studies have effectively demonstrated their performance for medical image processing. Motivated by these, we develop a deep-learning model to fully automatically and highly efficiently localize the lamina on transverse views of 3D spinal ultrasound volume data. The competitiveness of our model, called lamina localization network (LLNet), in lamina detection, comes from three hands. First, as shown in Fig. 1, the spinous process (SP) is a representation component of the “W” contour, and the SP and the lamina are located on the central peak and two-side valleys of the “W” contour, respectively. This is a valuable anatomical prior, and the SP is more observable and insensitive to image noise compared with the lamina. Inspired by this, we establish an anatomical prior embedding branch (SP-Prior) to implicitly mine the anatomical correlation between the SP and the lamina by inducing feature latent space alignment. We then exploit this correlation to optimize the rough localization result in inferring. Furthermore, many studies have indicated that information loss hurts tiny target localization caused by sequential convolutional and pooling operations, as demonstrated in Fig. 3, so we introduce two modules to overcome this problem. The first is to make full use of low-level feature maps, as they contain more details that can indicate the precise location of the target than the semantically information-rich high-level feature maps. Considering that our detection targets are two similar ROIs, there is a high correlation between these two targets and a low correlation with other non-target regions. Thus, we employ the AFMA module to refine the low-level feature maps and improve the detection precision by mining the correlation of left and right lamina in the low-level feature maps and corresponding to the original image. The second is utilized to optimize the features during propagation further. The SPM module is responsible for this, which constructs a set of parallel learnable parameters to capture contextual information by exploring the global-level long-range dependencies and the local-level consistency.

Our study makes three main contributions:

- (1) This study proposes an innovative vertebral anatomical prior guidance model, further combining two effective function modules to mitigate the negative effect of information loss due to down-sampling for small target detection, for fully automatic and highly efficient lamina detection in massive early screening and longitudinal follow-up. Our study is the first deep-learning model established for lamina localization.
- (2) Our model establishes a vertebral anatomical prior embedding branch to creatively employ the vertebral anatomical prior, by implicitly exploring the correlation between the SP and the lamina using feature latent space alignment. In inferring, the

correlation is exploited to highlight the potential region in which the lamina exists based on the prompt of SP position to enhance the model performance.

- (3) Cooperation with two functional modules to overcome the issue of information loss from both the perspective of exploiting low-level feature maps and optimizing feature propagation. The AFMA module provides refined low-level feature maps in which more details are highlighted that can indicate the accurate localization of laminae. The SPM module extracts contextual information by integrating global dependencies and local consistencies and then further optimizes features during feature propagation.

Subsequently, Section 2 summarizes recent representative studies, including scoliosis evaluation based on ultrasound, object detection approaches, and object detection approaches in medical images. Section 3 presents details of our model. Section 4 describes the involved data and the experimental settings. Results and discussion are given in Sections 5 and 6, respectively. Finally, the conclusion is displayed in Section 7.

2. Related works

2.1. Scoliosis evaluation based on ultrasound

Radiation-free and real-time superiorities contribute to the widespread employment of ultrasound in medical imaging. Fortunately, ultrasound imaging has been successfully introduced into clinical practices concerning scoliosis evaluation in recent years. Among numerous applications based on 3D spinal ultrasound imaging, the most representative was the volume projection imaging (VPI) technique (C Cheung et al., 2015). The VPI technique could project the entire 3D spinal ultrasound volume data onto a specified plane, such as coronal and sagittal. Then, the spinal anatomy can be visualized without radiation hazards. Similar to the Cobb angle measurement on the X-ray images, a quantification of scoliosis level can also be achieved on VPI images (ultrasound curve angle, UCA), and a high correlation between Cobb and UCA has been demonstrated effectively (Wong et al., 2019). Furthermore, some critical vertebral anatomies can still be observed in the VPI image, although the bone surface strongly reflects ultrasound waves. The transverse process and spinous process have the potential to describe the spinal profile (Zhou et al., 2020), the lamina is an essential landmark for assessing the vertebral rotation (Lee et al., 2023), and the rib and lump are observable in VPI images (Huang et al., 2022; Banerjee et al., 2024). Recently, the prediction of scoliosis progression was promisingly achieved by 3D spinal ultrasound imaging, which effectively avoids radiation hazards in longitudinal follow-up compared with

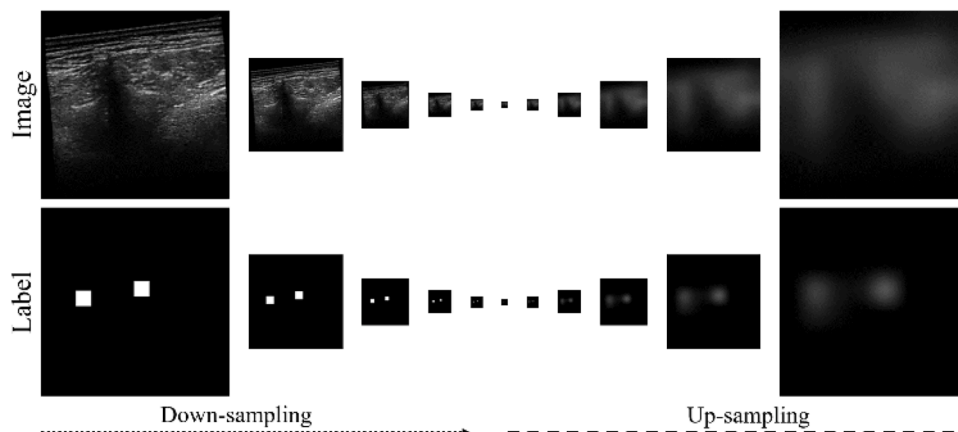


Fig. 3. Schematic illustration of information loss due to successive convolution and pooling.

radiographic scanning (Lai et al., 2024).

2.2. Object detection approaches

Region-proposal-based (RPB) frameworks and you-only-look-once (YOLO) series are two mainstreams in object detection tasks. RPB ones handle the task using a two-stage implementation, including bounding box acquiring and class probability calculating. Region-convolutional neural network (R-CNN) was the root of RPB frameworks. The R-CNN combined a CNN and a support vector machine (SVM) into a framework where the two components performed feature extraction for region proposal and classification (Girshick et al., 2014). Then, two representative improved versions of R-CNN were proposed, namely fast-R-CNN and faster-R-CNN, to overcome the limitation of low-efficiency (Girshick, 2015; Ren et al., 2015). Unlike RPB frameworks, the YOLO series simplified the task as a regression problem to establish a one-stage end-to-end architecture. Thus, the YOLO series were faster than the RPB methods, but their detection accuracy was sometimes lower than the latter. Due to the high detection efficiency, the YOLO series has been updated since its release in 2016 (Redmon et al., 2015), and several successful versions have been released during this period (Redmon and Farhadi, 2016; J Redmon and Farhadi, 2018; J Redmon and Farhadi, 2018). Apart from these classic detectors, there were also many outstanding detection models based on improved encoder-decoder architecture. Parallely, with the explosion of foundation models, some have also been successfully applied to object detection tasks through few-shot learning or necessary prompts (Jiao et al., 2023).

Object detection is a fundamental and critical task in medical image analysis. Compared with object detection tasks in natural images, those in medical images face more significant challenges due to inter-individual variability and intra-individual ambiguity, as well as the requirement for higher localization accuracy for clinical practices. Facing these challenges, numerous outstanding approaches have been proposed and performed well in this field. There are still plenty of YOLO- or RPB-based methods (Chang et al., 2024; Dosovitskiy et al., 2020; Su et al., 2022; Tarimo et al., 2024; Khan et al., 2023; Xu et al., 2023), in which the improvements were brought by effective preprocessing strategies, multi-scale feature refinement, and global dependency exploration. In addition to these models established based on YOLO or RPB architecture, some promising detectors performed well in medical imaging using an improved encoder-decoder one. The success of them indicated that enhancing multi-scale feature extraction capabilities, exploring into local consistency and global dependencies, introducing prior knowledge, and effective learning strategies, such as reinforcement learning, are beneficial to improve model performance (Wang et al., 2021; Y Wang et al., 2023; Tao et al., 2023; Wan et al., 2023).

2.3. Small objection detection approaches

In the small object detection task, most of the targets to be detected are scattered throughout the image, which makes it difficult for the model to accurately capture the exact location of all targets on a feature map at a certain scale; meanwhile, the information loss caused by successive convolution and down-sampling is not conducive to recovering the exact location of the small targets in decoding. To overcome these problems, researchers have done a lot of work and achieved promising results. Improving the multi-scale feature extraction capability of a model by diversifying the receptive field and then fusing extracted features is an effective solution for the first problem (Ma et al., 2023; Tong et al., 2023; T Zhang et al., 2023; T Wu et al., 2023), and there is even a successful practice of sub-model nesting to implement this method (X Wu et al., 2023). In addition to strengthening the feature extraction capability of the model, introducing more sources of information and thus acquiring more features can also solve this problem. Prior knowledge highlights certain properties of an image and is therefore a high-quality supplement of information. Spatial-temporal

(frequency) representation and image filtering approaches can provide more descriptions of an image from different perspectives, and the model can detect small targets, including low-quality images, more efficiently by fusing this information (Luo et al., 2023; Z Zhang et al., 2023). For the issue of information loss, a simple skip connection is incapable of mitigating it effectively, so researchers were committed to further enriching network connections and refining feature propagation. (Tian et al., 2023) and (Li et al., 2023) revealed that dense and nested connection strategies contributed an efficient feature communication between the encoder and the decoder.

These optimized strategies are also suitable for small object detection in medical imaging, including enhancing the feature extraction capability and optimizing the feature propagation (Li et al., 2024; Chen et al., 2023; Chen et al., 2024; Ghamsarian et al., 2022). More promisingly, a similar anatomical structure exists among different individuals, so the introduction of anatomical prior can be helpful for better small target detection in medical images. Some methods explicitly exploit prior knowledge to indicate regions where the target may be present, thus improving detection accuracy (Astaraki et al., 2022), while others implicitly integrate prior knowledge through feature representation and fusion to optimize detection results (Chen and Peng, 2024; He et al., 2024).

3. Methodology

3.1. Overview

The crucial motivation of our model is to utilize the vertebral anatomical prior to assist the lamina detection. The relative position between the SP and the lamina in the vertebral body is relatively stable. Meanwhile, the SP, unlike the lamina, is not severely affected by vertebral rotation in terms of its observability. Taking these two points into consideration, we constructed the SP-Prior branch to implicitly mine and construct the anatomical correlation between the SP and the lamina at the feature level by guiding the feature latent space alignment of the SP and the lamina, and employ this correlation to optimize the rough detection results in inferring. Parallely, considering the negative impact of information loss due to successive convolution and pooling in the small target detection, two other modules are introduced to overcome this problem. The first one is to make full use of the low-level feature map because it contains more details that can indicate the specific location of the target compared to the high-level feature map; the second one is to optimize the feature propagation and mine more useful information in this process to improve the detection accuracy. The AFMA module and the SPM module play important roles in achieving these two aims, respectively. Since our detection targets are two separated ROIs, it is conceivable that two similar targets have a strong correlation with each other and show a low correlation with non-similar parts. The AFMA module takes advantage of this to highlight the correlation between left and right lamina in the low-level feature maps and correspond them to the original image, aiming to optimize the low-level feature maps and optimize the rough detection results in decoding. The SPM module is deployed between the encoder and the decoder to improve feature propagation, which constructs a set of learnable parameters to extract contextual information by combining global long-range dependencies and local consistencies, and then improve the detection accuracy.

The LLNet is shown in Fig. 4, and it can be found that two parallel components consist of it. The main encoder-decoder with multi-scale feature extraction and skip connections is responsible for localizing the lamina preliminarily. Moreover, the AFMA and the SPM module cooperate with it to mitigate the information loss and improve the detection accuracy by refining low-level feature maps and extracting contextual information to optimize feature propagation. More valuably, a vertebral anatomical prior embedding branch avoids the drawback of laminal poor observability caused by vertebral rotation and utilizes the

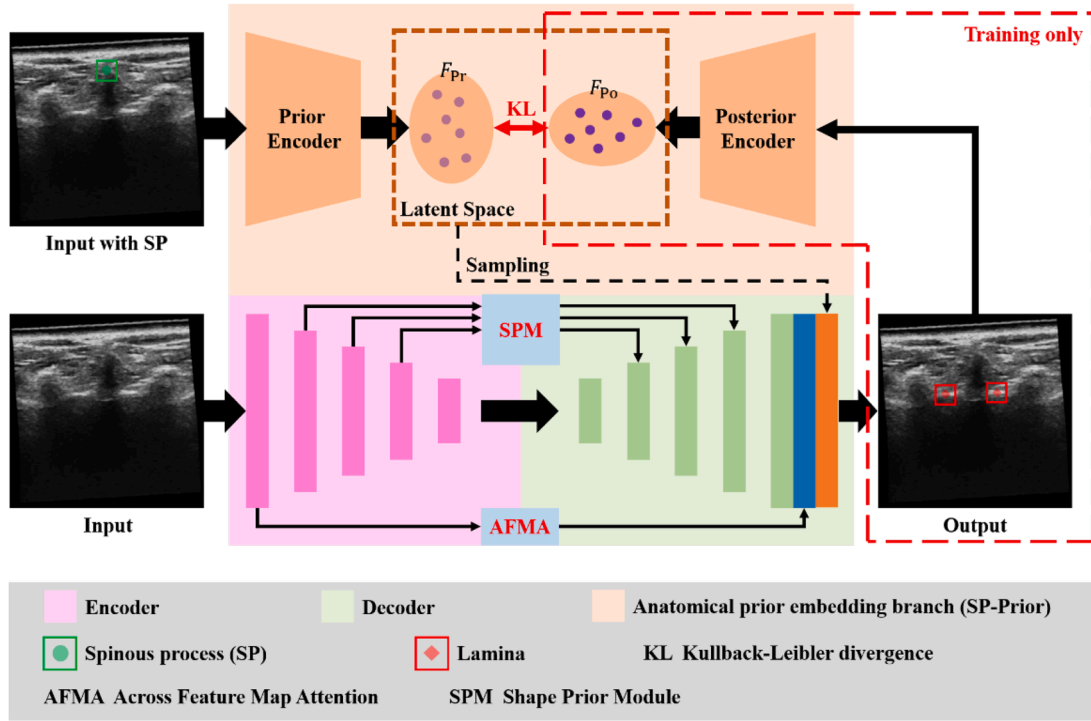


Fig. 4. Framework of the lamina detector named LLNet. An anatomical prior embedding branch runs parallel to a main encoder-decoder architecture. A prior and a posterior encoder establish a joint latent space based on the concatenation between the slice and the SP annotation and that between the slice and the lamina label, respectively. Then, KL penalizes differences between prior (F_{Pr}) and posterior (F_{Po}) features to correlate the position of SP and that of lamina, and SPM further explore the contextual information and the local consistency. Finally, the rough detection result is optimized by the AFMA-refined low-level features and the anatomical prior features sampled from the latent space.10.

more observable SP as an agent to implicitly mine and further employ the anatomical correlation between the SP and the lamina to highlight the potential region in which the laminae exist.

3.2. Anatomical prior embedding branch (SP-Prior)

Many studies have demonstrated that the introduction of prior knowledge can effectively improve model performance, especially in medical image analysis, where the anatomical structure is a very valuable prior knowledge because it is broadly stable across various individuals and therefore widely applicable (Hu et al., 2023; Zhang et al., 2024). In our study, the vertebral observability tends to decrease due to vertebral rotation. Thus, direct employment of anatomical prior about lamina is difficult and so requires an agent. Fortunately, another critical landmark on the vertebra (SP) has relatively stable observability and, as shown in Fig. 1, the relative positional relationship between the SP and the laminae is relatively fixed, making it an ideal agent for indicating the potential region in which the laminae exist. Taking these factors into account, we designed the SP-Prior branch to utilize the vertebral anatomical prior. Motivated by the feature latent space alignment strategy (Liu et al., 2023; Patel and Dolz, 2022; P Wang et al., 2023; Zhai et al., 2023), the SP-Prior branch constructs two encoders for extracting features related to the SP and the lamina with the help of the corresponding guided mask, respectively. Subsequently, the anatomical correlation between the SP and the lamina is implicitly mined and constructed at the feature level by guiding the feature latent space alignment of the SP and the lamina. Finally, improved features with the anatomical correlations are sampled from the aligned latent space to optimize the rough detection results in the inferring phase. The implement details of the SP-Prior branch is shown in Fig. 4.

Concretely, a prior and a posterior encoder with a similar architecture are constructed to extract features from the concatenation between the transverse slice and corresponding SP and lamina annotation,

respectively. The concatenation input brings two benefits, one is the local region of SP and lamina is highlighted by corresponding guided masks, which is conducive to encoding valuable local information, and the other one is the encoder also can capture global information of the entire image since the original image serves as a part of the input. In this case, the feature latent space can contain both the global information of entire image and the local one related to the SP and the lamina, which facilitates the mining of the anatomical correlation between the SP and the lamina. Next, the correlation between them needs to be constructed. We define that F_{Pr} and F_{Po} are feature distributions respectively acquired from the prior and the posterior encoder in the joint latent space, and a sample from F_{Pr} and F_{Po} are respectively described as:

$$s_i \sim F_{Pr}(\cdot|\mathbf{X}) = \mathcal{N}(\mu_{Pr}(\mathbf{X}), \text{diag}(\sigma_{Pr}(\mathbf{X}))), \quad (1)$$

$$z_i \sim F_{Po}(\cdot|\mathbf{X}, \mathbf{Y}) = \mathcal{N}(\mu_{Po}(\mathbf{X}, \mathbf{Y}), \text{diag}(\sigma_{Po}(\mathbf{X}, \mathbf{Y}))), \quad (2)$$

in which $\mathbf{X}$ and $\mathbf{Y}$ means the encoder input and the ground truth, respectively. The mean and the variance of F_{Po} are denoted as μ_{Po} and σ_{Po} , respectively. Similarly, the F_{Pr} has the mean μ_{Pr} and the variance σ_{Pr} . Through aligning these two distributions in the feature latent space, the anatomical correlation between the SP and the lamina can be implicitly explored. Aiming to this, a cross-entropy loss is used to promote the similarity between the output and the ground truth. Meanwhile, a sample from the F_{Po} , due to having the precise local information concerning the lamina, can optimize the rough detection result effectively for the training example, so a Kullback-Leibler (KL) divergence is further employed to penalize the difference between F_{Pr} and F_{Po} , as defined in (3) and (4):

$$\mathcal{L}(\mathbf{Y}, \mathbf{X}) = \mathbb{E}_{z_i \sim F_{Po}(\cdot|\mathbf{Y}, \mathbf{X})} [-\log F(\mathbf{Y}|D_{Final}(\mathbf{X}, z))] + \beta KL(F_{Po}(z|\mathbf{Y}, \mathbf{X}) \| F_{Pr}(z|\mathbf{X})), \quad (3)$$

$$KL(F_{Po}(z|\mathbf{Y}, \mathbf{X}) \| F_{Pr}(z|\mathbf{X})) = \mathbb{E}_{z_i \sim F_{Po}} [\log F_{Po}(z|\mathbf{Y}, \mathbf{X}) - \log F_{Pr}(z|\mathbf{X})]. \quad (4)$$

F is a pixel-wise parameterized categorical distribution for the implementation of cross-entropy loss calculation, and D_{final} in the final output. In the training phase, the cross-entropy loss and the KL penalization drive the F_{pr} and the F_{po} closer to each other, and the F_{pr} gradually become able to describe all presented variants of anatomical correlation between the SP and the lamina. Thus, in the inferring phase, the posterior encoder is no longer used, and the aligned prior encoder estimates the affinity between a new input with its SP mask and encoded variants and then provides features which can highlight the potential region of lamina based on the anatomical correlation to further improve the rough result. Concretely, the sample s_i with anatomical prior is concatenated with the rough output of decoder, and then they are processed by continuous 1×1 convolutional units ($Conv_{1 \times 1}$) to become the final output, which is summarized as:

$$D_{final} = Conv_{1 \times 1}(Cat(D_r, s_i)) \quad (5)$$

in which $Cat(*)$ represents the concatenation between components in $*$, and D_r is the rough detection result.

Additionally, the annotation of SP used in the anatomical prior embedding branch is automatic, and the algorithm is achieved based on the distinctive intensity distribution of the transverse slice. As mentioned in Fig. 1, the “W”-shape of the vertebra is reflected in the transverse slice and the intensity histogram along the direction of ultrasound wave injection. Motivated by this, we design a twice-histogram-localization algorithm to annotate the SP automatically, as displayed in Fig. 5. First, we localize the potential region around the rough central valley of the “W” contour in the horizontal intensity histogram and an image patch with a width of 100 pixels. Then, the height of the image patch is decided by the low-intensity region of the vertical histogram. Second, the extracted image patch is binarized, and we calculate the horizontal intensity histogram of it. Similarly, the central valley of the “W” contour is localized in the fine-scale horizontal intensity histogram, and the contact with the binary image patch is defined as the SP automatic annotation. In this way, our model can be fully automatic.

3.3. Across feature map attention (AFMA)

Successive convolution and pooling operations inevitably cause information loss, which is one of the motivations for establishing skip, residual, and dense connections. Further, this information loss is also detrimental to small target detection. In our study, since spine imaging

mainly relies on the strong reflection of ultrasound waves from bone surfaces, the vertebra, as one of the vertebral structures, tends to show similar grey-scale distribution properties as other adjacent bone structures. In this case, it is difficult to accurately locate the vertebra only based on high-level feature maps that are rich in semantic information but lacking in spatial information. Accordingly, we are counting on further exploring and utilizing low-level feature maps rich in spatial information to achieve accurate vertebra detection. Meanwhile, our task aims to detect two targets in the same category, so theoretically, there should be a high similarity between these two targets and a low similarity with others. Taking these two factors into consideration, we introduced the AFMA module in our model (Sang et al., 2022). The AFMA module constructs localization guidance by mining the similarities of targets belonging to the same category and then mapping them to the original image and then optimizes the rough detection result with the guidance in decoding.

Concretely, the input image is successively encoded by 7×7 convolutions with 64 filters and 3×3 one with 1 filter as can be seen in Fig. 6, and acquired low-level features (F_{img}) is split into fixed-size patches (P_{img}) by a separation operation (Sep). In this case, the F_{img} is reorganized as a series of flattened two-dimensional patches P_{img} , and the resolution of j -th vector of P_{img} is $r \times r$. Then, one 1×1 convolution with N_c (the number of prediction categories) filters convert the i th level feature maps (F_{ith}) to per category features, so that the similarity between targets belonging to the same category can be highlighted in the corresponding feature map. Similarly, each channel of F_{ith} is also separated into a sequence of flattened patches (P_{ith}), respectively, and then these patches are concatenated together. Finally, the dot product between the F_{img} and the transposed version of each P_{ith}^k is performed, which aims to explore the similarity between targets belonging to the same category and correspond that into image patches of the original image and the sequence of flattened patches. The implementation of AFMA can be summarized as:

$$P_{img} = Sep(F_{img}, r), \quad (6)$$

$$P_{ith}^k = Sep(F_{ith}^k, r), P_{ith} = \mathbb{C}(P_{ith}^1, \dots, P_{ith}^{N_c}), \quad (7)$$

$$A_{ith}^k = P_{img} \otimes (P_{ith}^k)^{-1}, A_{ith} = \mathbb{C}(A_{ith}^1, \dots, A_{ith}^{N_c}), \quad (8)$$

in which $\mathbb{C}(*)$ means the concatenation of members in $*$. Subsequently, the attention map A_{ith} provided by the AFMA is further used to improve

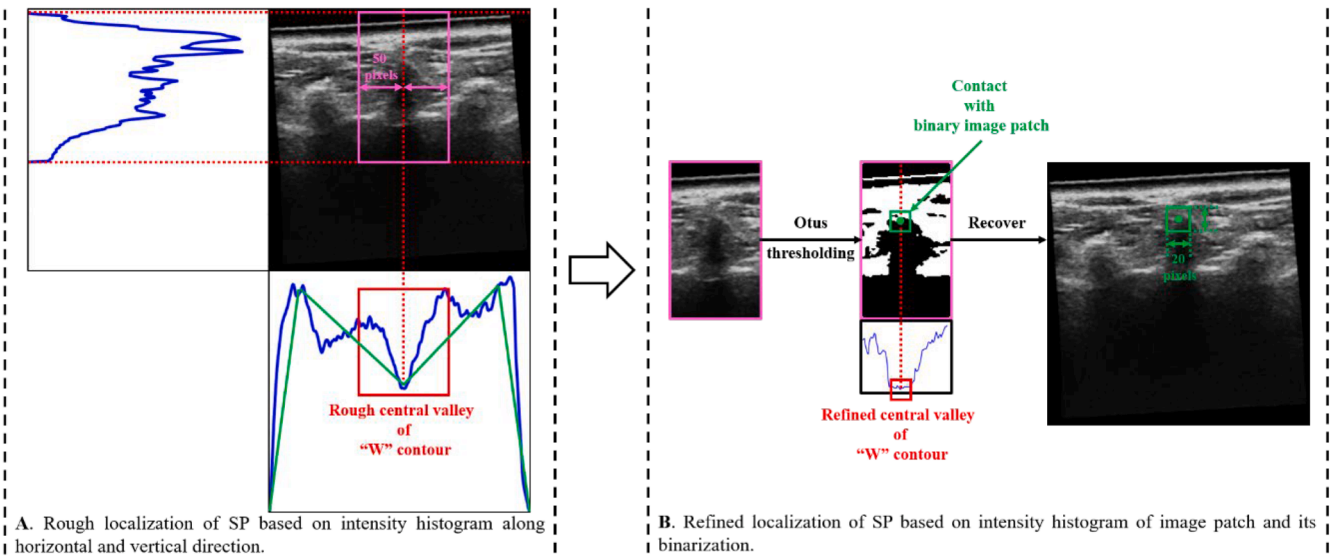


Fig. 5. Implementation of automatic annotation of spinous process (SP) in transverse slice based on twice-histogram-localization algorithm.

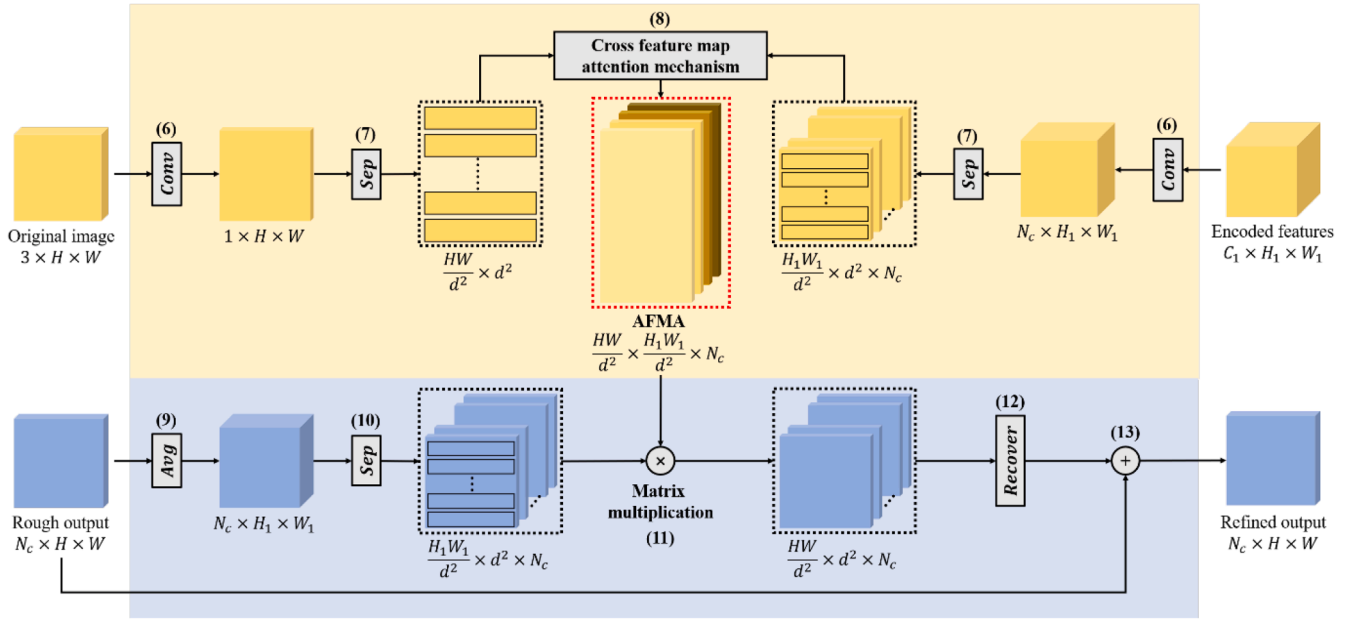


Fig. 6. Architecture of across feature map attention (AFMA). Low-level features without serious information loss is correlated with the original image via a cross feature map attention mechanism, and then acquired AFMA is used to further refined the rough result.

the rough detection result (D_r) without serious information loss. An average pooling operation (Avg) with kernel size of (k_H, k_W) and stride of (s_H, s_W) compresses the D_r to the same size as the F_{ith} , and then compressed results of all categories are concatenated, too (D_c). Parallely, each channel of D_c is also separated and concatenated, and then the attention map A_{ith} can be utilized to optimize the corresponding patch of rough detection result. Finally, the optimized patches are recovered to the same size as the D_r to achieve the detection improvement. The optimization process can be described as:

$$D^k = \text{Avg}(D_r^k), D = \mathbb{C}(D^1, \dots, D^{N_c}), \quad (9)$$

$$D_s^k = \text{Sep}(D^k, r), D_s = \mathbb{C}(D_s^1, \dots, D_s^{N_c}), \quad (10)$$

$$D_{ith}^k = A_{ith}^k \otimes D_s^k, D_{ith} = \mathbb{C}(D_{ith}^1, \dots, D_{ith}^{N_c}), \quad (11)$$

$$\bar{D}_{ith}^k = \text{Sep}^{-1}(D_{ith}^k, r), \bar{D}_{ith} = \mathbb{C}(\bar{D}_{ith}^1, \dots, \bar{D}_{ith}^{N_c}), \quad (12)$$

$$D_{Final} = D_r + \bar{D}_{ith}. \quad (13)$$

Sep^{-1} represents the folding operation as the inverse one of Sep , and $\bar{D}_{ith}$ indicates the improved i th level feature maps. D_{Final} is the final output with the assistance of AFMA.

3.4. Shape prior module (SPM)

As mentioned before, information loss is detrimental to small target detection. In addition to effectively utilizing low-level feature maps as in the AFMA module, we have also optimized feature propagation. Skip connections have been widely demonstrated to be effective as a classical method to overcome this issue. In our model, this design is also employed, and we further modify the skip connections to extract more favorable information to assist in lamina detection. Transformer-based models, such as DETR (Carion et al., 2020), have revealed that global dependencies brought by the self-attention mechanism are valuable for improving model performance. Thus, we introduce the SPM module to exploit global dependencies in our model and improve detection accuracy further. The SPM module constructs a set of learnable prior to mine global dependencies based on a self-attention mechanism, which also raises the similarity between the two targets belonging to the same

category like the AFMA module does, and in parallel, it further cooperates with a convolutional sub-module with local consistencies, to refine skipped features (You et al., 2023).

The architecture of SPM is visualized in Fig. 7. It can be found that a cross-update block (CUB) and a self-update one (SUB) mainly consist of the SPM, and the CUB and the SUB are responsible for retaining the local consistencies and mining the global dependencies, respectively. In the CUB, the learnable priors of image pattern F_{learn} is firstly up-sampled as the same resolution of skipped features (F_{skip}) to ensure the interactivity between them, and then they are integrated by the cross-attention mechanism (Carion et al., 2020). The affinity map (A_{fc}) between them can be defined as:

$$A_{fc} = \text{Softmax} \left(\frac{Q_c(F_{skip}) \times K_c(\text{Up}(F_{learn}))^T}{\sqrt{N}} \right), \quad (14)$$

$$\bar{F}_{skip} = A_{fc} \times V_c(\text{Up}(F_{learn})) + F_{skip}, \quad (15)$$

$Q_c(\cdot)$, $K_c(\cdot)$ and $V_c(\cdot)$ transform $\cdot$ into the query, the key and the value in the CUB, respectively. Up and T correspond to the up-sampling and transpose operation, respectively. N is the channel number of priors. Then, the F_{skip} is further improved according to the A_{fc} with embedding the F_{learn} , as demonstrated in (15). F'_{learn} represents the optimized F_{learn} provided by the SUB. The exploration of global dependencies is performed in the SUB. The first step in the SUB is similar to that in the CUB, to calculate the affinity (A_{fs}) among channels of learnable priors as:

$$A_{fs} = \text{Softmax} \left(\frac{Q_s(F_{learn}) \times K_s(F_{learn})^T}{\sqrt{N}} \right), \quad (16)$$

$$F'_{learn} = \text{LayerNorm}_1(A_{fs} \times V_s(F_{learn})), \quad (17)$$

$$F'_{learn} = \text{LayerNorm}_2(\text{MLP}(F'_{learn})). \quad (18)$$

$$\bar{F}_{learn} = \text{Down}(\bar{F}_{skip}) + F'_{learn}. \quad (19)$$

$Q_s(\cdot)$, $K_s(\cdot)$ and $V_s(\cdot)$ transform $\cdot$ into the query, the key and the value in the SUB, respectively. After that, the weighted sum of A_{fs} and vectorized F_{learn} are combined to mine the global dependencies, which is implemented by the cooperation between a multi-layer perceptron

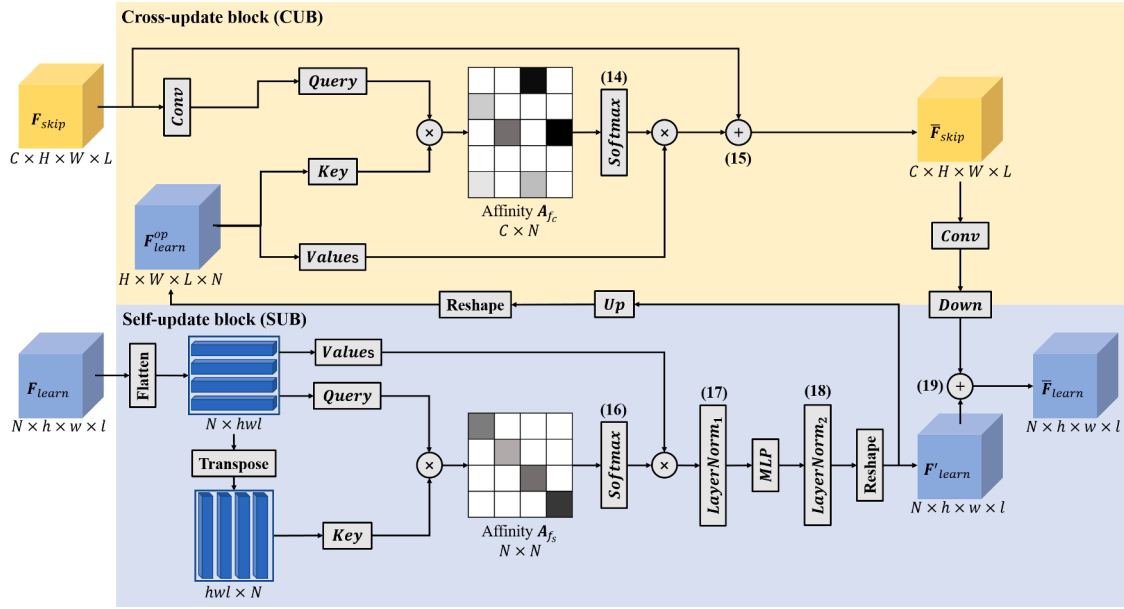


Fig. 7. Architecture of shape prior module. Cross-update block explores the local consistencies with the assistance of learnable prior provided by self-update block. Self-update block contributes to summarize the global dependencies into the learnable prior with the refined skipped features. Their combination effectively extract global dependencies and local consistencies so improve the model performance.

(MLP) and layer normalization (LayerNorm) units and illustrated as (17) and (18). The improved $\bar{F}_{skip}$ is subsequently injected into the SUB, to parallelly refine the learnable priors with the inductive bias of locality and scale invariance, which is summarized as (19). Finally, $\bar{F}_{skip}$ and $\bar{F}_{learn}$, the refined skip feature and the updated learnable prior from the SUB and the CUB, are obtained with sufficient global information and local bias.

4. Materials and experiments

4.1. Materials

A total of 150 subjects were included in our study. All 3D spinal ultrasound volume data were obtained in actual clinical practice, the equipment used for data acquisition was Scolioscan® system, and the operations were performed by clinical experts with the assistance of engineers. The number of transverse slices in each volume of data ranges from 2220 to 3392, with the average containing 2942 slices. Subsequently, 100 slices were randomly selected from each subject to constitute our data, totaling 15,000 slices, and data from 100, 10, and 40 subjects were randomly organized as training, validating, and testing dataset, respectively, which means that the training, validating and testing dataset consist of 10,000, 1000 and 4000 slices, respectively. Four experienced researchers conducted the manual annotation of lamina under the expert's guidance, and the annotation was repeated three times to minimize inter-observer and intra-observer error. Finally, the expert fine-tuned the annotation and selected the final ground truth.

4.2. Experiments

Our experiments contained two parts. First, a comparison experiment was performed with some outstanding models in localizing small targets and classical detectors, including UIUNet, DeepPyramid, DNA-Net, AGPCNet, MTUNet, MambaUNet, YOLO v7, and Faster R-CNN, to highlight the superiority of our model in localizing the lamina. UIUNet is specially designed for small target detection, and it enables multi-level and multi-scale representation learning of small objects, improving small target detection accuracy (X Wu et al., 2023). DeepPyramid also performs well for small target detection via introducing a deformable

pyramid reception module to provide a wider receptive field and extract more multi-scale features (Ghamsarian et al., 2022). DNANet overcomes the problem of information loss via a progressive and effective interaction among high-level and low-level features achieved by a dense nested interactive module (Z Zhang et al., 2023). Unlike the above approaches that improve the capability in multi-scale feature extraction and fusion for accurate small target detection, AGPCNet and MTUNet promote a precise small target localization by mining and integrating global dependencies within and between patches at specific scales by an attention-guided pyramid module which exploits and fuses local semantic association and global context attention (T Zhang et al., 2023). MTUNet employs a hybrid encoder consisting of convolution layers and vision transformer ones to extract and integrate local consistencies and global dependencies (T Wu et al., 2023). Similarly, MambaUNet performs well in several medical image segmentation tasks, containing small polyp and skin lesion segmentation since the Mamba module brings a powerful global context information extraction capability (Wang et al., 2024). YOLO v7 and Faster R-CNN are two classical detectors, which have been widely employed in target detection tasks and performed well. Second, comprehensive ablation studies were used to investigate the effectiveness and contribution of innovative components in the LLNet. The original encoder-decoder architecture is defined as the backbone. Then, innovations are gradually combined with the backbone to improve the model performance, including the SP-Prior module, the SPM one, and the AAFM one. Moreover, the backbone is further combined with two innovations, aiming to investigate the contribution of the integration between functional modules. In addition to the widely used detection speed (frames per second, fps) and intersection over union (IoU) as evaluation metrics, we further defined detection success rate (Sr) as an auxiliary metric. This is because a 3D spinal pose descriptor is subsequently constructed based on the detected results, some results with IoU greater than 0 but still at a considerable distance from the corresponding ground truths, which tends to increase the difficulty of post-processing for constructing a 3D spinal pose descriptor so should be discarded. Motivated by this, the detection success rate at different thresholds is employed, to further reveal the potential of our model to be used in clinical practice in the future. Their calculation are defined as follows:

$$Sr = (N_s/N_{All}) \times 100\%, \quad (20)$$

$$IoU = (R_D \cap R_G)/(R_D \cup R_G), \quad (21)$$

The definition of success detection is demonstrated in Fig. 8. If the bounding box of detection result has an intersection with that of ground truth, the detection will be considered as success, and the criteria of successful detection gradually become more stringent with the increment of threshold. In (1), N_s and N_{All} is the number of slices in which the lamina detection is achieved and the number of all slices, respectively. Due to negative factors such as vertebral rotation, the observability of the left and right lamina varies. Therefore, in addition to assessing the overall detection success rate, the assessment was performed for the left and right targets, respectively, using criteria consistent with the definitions in Fig. 8. The detection speed is defined as the number of slices for the lamina detection in one second, namely frames per second (fps). The IoU is introduced to evaluate the detection accuracy, and a larger IoU indicates a greater intersection between the detected result and the ground truth. R_D and R_G means the region of detected bounding box and that of ground truth, respectively.

4.3. Implementation details

The loss function of LLNet is composed of two components. One is the KL divergence defined in (2) to align the SP prior and the lamina posterior. The other is a cross-entropy loss calculated between the detection results and the ground truth. Our model in this study was implemented based on the Pytorch framework running on an NVIDIA® Tesla V100-SXM2. The Adam optimizer was introduced to update the model parameters with an initial learning rate $3e-3$ and a linear decay strategy. All slices were resized to $256 \text{ pixels} \times 256 \text{ pixels}$ and then fed into the model. Batch size and the number of epochs were set to 128 and 1000, respectively. The setting of other compared models was similar to the above. The model that performs best on the validating dataset was utilized for further evaluation and comparison on the testing dataset. Finally, MATLAB® 2018a (MathWorks, Natick, MA, USA), running on an Intel® Xeon® Silver 4215 CPU at 2.50 GHz, was used for data processing and analyzing.

5. Results

5.1. Performance comparison

Eight outstanding models were introduced in our comparison experiment, including UIUNet, DeepPyramid, DNANet, AGPCNet, MTUNet, MambaUNet, YOLO v7, and Faster R-CNN. On the one hand, Table 1 shows these models' detection success rate. It can be seen that the LLNet achieved an effective lamina detection, as revealed by a 100.0 % detection success rate. Further, the LLNet also obtained a 100.0 % detection success rate in the independent evaluation of the left and right lamina, respectively. Parallely, the detection success rate of the LLNet still has a satisfactory competitiveness compared with other models as

the increment of overlap ratio, as demonstrated in Table 1 and Fig. 9. On the other hand, the criteria of success detection become more stringent as the threshold increases so that it can be found in Tables 1 that the detection success rate of other detection models showed varying degrees of degradation. By contrast, the LLNet was not seriously affected by threshold changes and showed stable performance. Moreover, as displayed in Table 2, the best mean IoU with the value of 0.86 is also obtained by the LLNet, which has an 8.9 % increase compared with other detection models and further highlights our model's competitiveness. For the detection efficiency, the highest detection speed was given by the YOLOv7, about 225 fps. Compared with it, the detection speed of the LLNet is about 20 fps. Fortunately, our model can still achieve quasi-real-time detection if deployed for use since the imaging speed of the Scolioscan is about 24 fps. Except for quantitative evaluation, some visualization examples are given in Fig. 10. As shown in Fig. 10, the region-of-interest (ROI) automatically delineated by the LLNet almost covered its ground truth. Even for those images with poor imaging quality, the LLNet provided more accurate lamina localization results than other detectors. Overall, compared to optimization strategies such as enhancing the model capability in extracting multi-scale features, such as UIUNet, DNANet and DeepPyramid, or integrating global dependencies and local consistencies, like AGPCNet, MTUNet and MambaUNet, mining and exploiting anatomical correlations are more likely to improve the model performance, which is precisely the strategy the LLNet implemented.

5.2. Ablation study

5.2.1. Effectiveness of SP-Prior

Ultrasound imaging takes the advantages of radiation-free and real-time imaging but also suffers from the disadvantages of being susceptible to interference and poor image quality. These determinations will likely pose a significant challenge for detecting small targets in ultrasound medical images. In this study, our model established a parallel anatomical prior embedding branch, namely the SP-Prior module, to explore an anatomical correlation between the target and another obvious anatomy and then utilized the correlation to indicate the potential region in which the lamina exists. As demonstrated in Table 3, with the assistance of the SP-Prior module, the detector achieved a 5.8 % improvement in detection success rate, and the detection capability became more stable in stricter thresholds. In the independent detection success rate assessment for left and right lamina, the SP-Prior still contributes to a more stable detection capability, as shown in Fig. 11. Parallely, an 8.6 % improvement in mean IoU was also obtained compared with the backbone, as displayed in Table 4. Similar conclusions can be drawn based on visualization examples in Fig. 12. Comparing the examples belonging to the backbone and the backbone w SP-Prior, the localization boxes given by the latter were more accurate than the counterparts from the former.

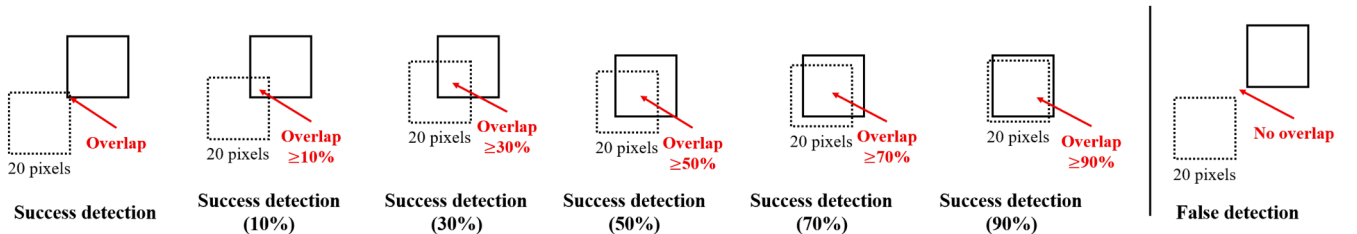
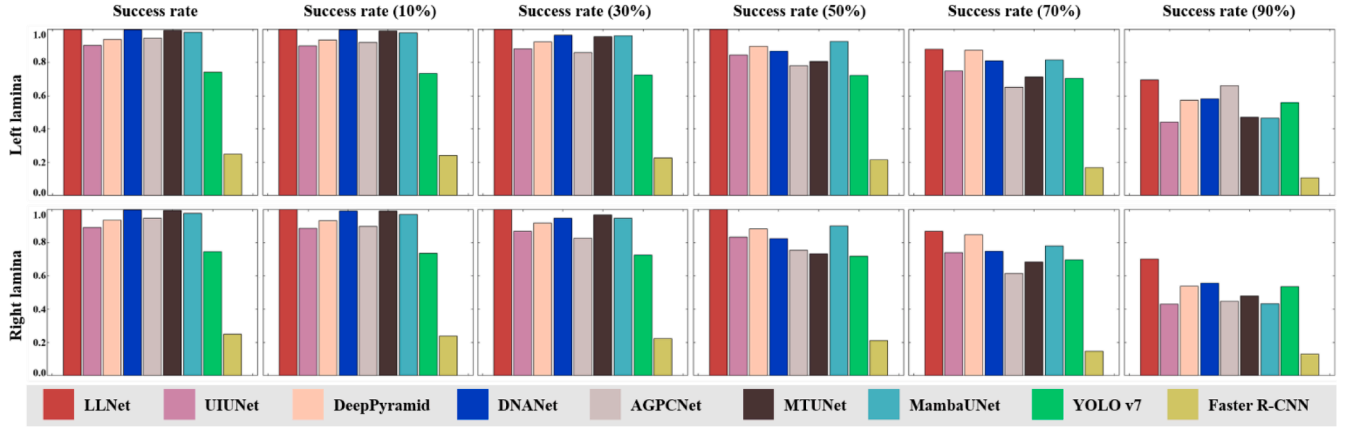


Fig. 8. Definition of success and false detection with different thresholds for a comprehensive evaluation of detection success rate, and the threshold is defined by the ratio between the intersection area to the area of the bounding box. Dashed and solid bounding boxes mean the detection result and the ground truth, respectively, and the detection is considered as successful when there is an intersection between the two. As the threshold increases, the criteria for successful detection become more stringent. If the two boxes do not intersect, the detection is considered to have failed.

Table 1

Overall detection success rate in the comparison experiment. Success rate (*) means the success rate under the overlap ratio of * between the detection result and the ground truth. Entries in bold and underlining highlight the best one and the second one, respectively.

Model	LLNet	UIUNet	DeepPyramid	DNANet	AGPCNet	MTUNet	MambaUNet	YOLO v7	Faster R-CNN
Left-right laminae									
Success rate	100.0 %	88.1 %	92.9 %	96.7 %	89.1 %	95.7 %	<u>96.6 %</u>	73.0 %	25.8 %
Success rate (10 %)	100.0 %	87.1 %	92.2 %	89.7 %	84.2 %	89.6 %	<u>95.7 %</u>	72.3 %	22.8 %
Success rate (30 %)	100.0 %	84.2 %	90.0 %	86.3 %	80.1 %	86.8 %	<u>92.3 %</u>	71.9 %	21.1 %
Success rate (50 %)	100.0 %	78.4 %	85.3 %	83.4 %	67.2 %	77.9 %	<u>85.7 %</u>	71.1 %	20.0 %
Success rate (70 %)	85.9 %	63.6 %	<u>77.6 %</u>	69.7 %	43.7 %	68.7 %	67.9 %	67.3 %	14.6 %
Success rate (90 %)	66.7 %	23.5 %	35.2 %	24.2 %	22.0 %	25.8 %	23.5 %	<u>43.1 %</u>	10.3 %

**Fig. 9.** Barplots of detection success rate for left and right lamina in the comparison under increasing overlap thresholds.**Table 2**

Evaluation metric summary in the comparison experiment. Entries in bold and underlining highlight the best one and the second one, respectively.

Model	LLNet	UIUNet	DeepPyramid	DNANet	AGPCNet	MTUNet	MambaUNet	YOLO v7	Faster R-CNN
IoU	0.86 ± 0.04	0.72 ± 0.13	0.73 ± 0.11	<u>0.79 ± 0.06</u>	0.71 ± 0.07	0.77 ± 0.06	0.77 ± 0.09	0.74 ± 0.24	0.61 ± 0.09
IoU (Left lamina)	0.87 ± 0.12	0.70 ± 0.15	0.71 ± 0.15	<u>0.77 ± 0.08</u>	0.73 ± 0.09	0.75 ± 0.08	0.75 ± 0.13	0.74 ± 0.24	0.63 ± 0.15
IoU (Right lamina)	0.86 ± 0.09	0.72 ± 0.17	0.71 ± 0.16	<u>0.75 ± 0.08</u>	0.75 ± 0.09	<u>0.79 ± 0.08</u>	0.75 ± 0.14	0.73 ± 0.24	0.60 ± 0.14
Speed (fps)	18.29	23.52	33.87	162.35	40.67	36.79	30.34	225.64	48.70

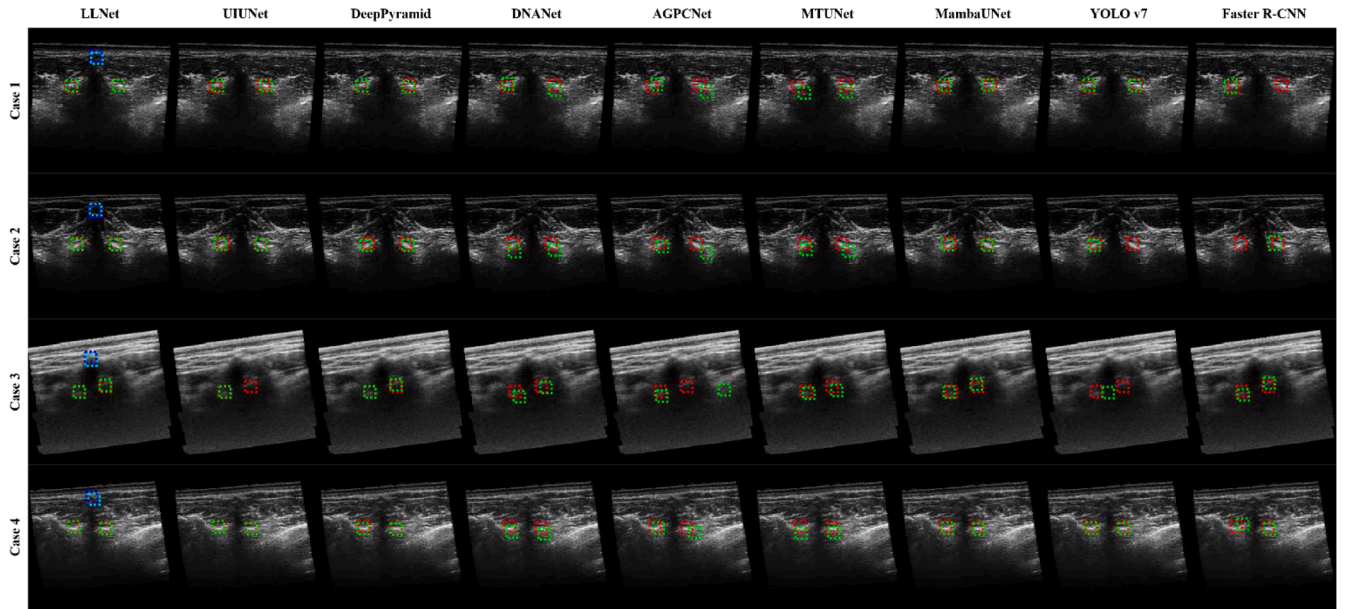
**Fig. 10.** Visualization examples of lamina detection in the comparison with other detection models. Green and red dashed bounding boxes represent the detected ROI containing laminae and corresponding ground truths, representatively. Additionally, the SPs detected by our searching algorithm and their corresponding annotations are visualized by blue and cyan dashed bounding boxes in the results of our model, respectively.

Table 3

Overall detection success rate in the ablation study. Success rate (*) means the success rate under the overlap ratio of * between the detection result and the ground truth. Backbone is symbolized as # and w * represents the backbone cooperates with * module.

Model	LLNet	Backbone	# w SP-Prior	# w AFMA	# w SPM	# w (SP-Prior & AFMA)	# w (SP-Prior & SPM)	# w (AFMA & SPM)
Left-right laminae								
Success rate	100.0 %	80.9 %	86.7 %	82.2 %	82.6 %	97.6 %	97.6 %	99.0 %
Success rate (10 %)	100.0 %	80.1 %	86.2 %	81.4 %	81.7 %	94.5 %	92.3 %	94.3 %
Success rate (30 %)	100.0 %	74.9 %	84.2 %	80.7 %	80.2 %	91.7 %	87.3 %	91.3 %
Success rate (50 %)	100.0 %	72.8 %	84.0 %	80.1 %	79.7 %	89.5 %	85.7 %	84.7 %
Success rate (70 %)	85.9 %	66.9 %	73.3 %	71.4 %	69.3 %	83.3 %	81.4 %	79.9 %
Success rate (90 %)	66.7 %	23.1 %	27.9 %	28.0 %	26.1 %	56.2 %	54.8 %	45.1 %

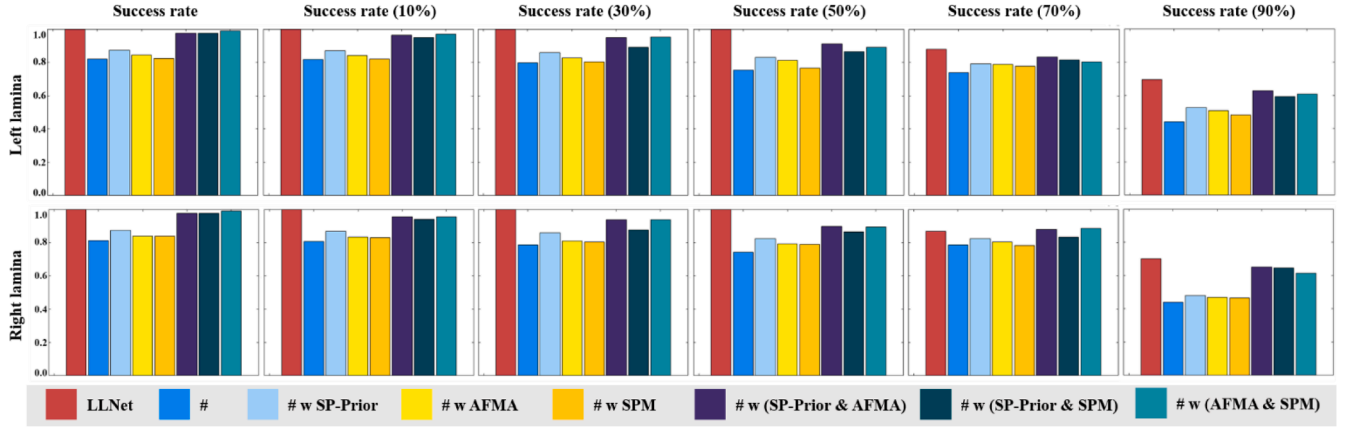


Fig. 11. Barplots of detection success rate for left and right lamina in the comparison under increasing overlap thresholds. Backbone is symbolized as # and w * represents the backbone cooperates with * module.

Table 4

Evaluation metric summary in the ablation study. Backbone is symbolized as # and w * represents the backbone cooperates with * module.

Model	LLNet	Backbone	# w SP-Prior	# w AFMA	# w SPM	# w (SP-Prior & AFMA)	# w (SP-Prior & SPM)	# w (AFMA & SPM)
IoU	0.86 ± 0.04	0.70 ± 0.12	0.76 ± 0.10	0.75 ± 0.10	0.73 ± 0.11	0.83 ± 0.12	0.81 ± 0.12	0.82 ± 0.12
IoU (Left lamina)	0.89 ± 0.06	0.67 ± 0.15	0.76 ± 0.14	0.74 ± 0.13	0.71 ± 0.14	0.84 ± 0.15	0.82 ± 0.18	0.83 ± 0.16
IoU (Right lamina)	0.90 ± 0.09	0.73 ± 0.16	0.78 ± 0.13	0.76 ± 0.14	0.75 ± 0.14	0.84 ± 0.16	0.82 ± 0.15	0.82 ± 0.16
Speed (fps)	18.29	32.15	30.27	21.33	25.60	19.34	20.17	19.96

5.2.2. Effectiveness of AFMA

The information loss caused by continuous convolutional and pooling operations also hurts detecting small targets. In our detector, the AFMA module was critical in mitigating the information loss and enhancing the model performance. The AFMA module refined low-level features based on the guidance of the original image and then directly reutilized the refined features without profound information loss to assist the final detection. The optimization of the detection success rate shown in Table 3 highlighted the beneficial role of the AFMA with a 1.3 % improvement. More importantly, the detection success rate became more stable with the threshold increment, as displayed in Fig. 11. The contribution of the AFMA was also indicated in Table 4. It can be found that the cooperation with the AFMA module satisfactorily brought the improvement, as the mean IoU were improved by 7.1 % compared with the backbone. Observing visualization examples of the backbone w AFMA in Fig. 12, its detection results are closer to their ground truth than those of the backbone, and the similar conclusion can be also obtained by examples in Fig. 13.

5.2.3. Effectiveness of SPM

Global dependencies and contextual information are valuable for constructing a representative image pattern. Motivated by this, the SPM module was introduced in our model. The SPM utilized the global attention mechanism to extract the global information update learnable priors and then polish multi-scale feature maps based on updated priors

to refine the local consistencies. With the assistance of the SPM, the lamina detection results provided by the backbone w SPM became closer to corresponding ground truths, as demonstrated by evaluation metrics in Tables 3 and 4. The detection success rate and mean IoU of the backbone w SPM increased by 1.7 % and 4.3 %, respectively. The effectiveness of SPM was also revealed by Fig. 11 in independent detection success rate analysis. In visualization examples of the backbone w SPM in Fig. 12, a more significant overlap can be observed between the detection results of the backbone w SPM and the gold standard compared to those of the backbone.

5.2.4. Effectiveness of inter-module cooperation

The inter-module cooperation brings a further improvement in the model performance. The collaboration between the SP-Prior and the AFMA module implicitly introduces anatomical prior knowledge. Further, it overcomes the problem of information loss based on the optimization and re-utilization of low-level feature maps. Similarly, the cooperation between the SP-Prior and the SPM module obtains better performance because the SPM module further mines and integrates global dependencies and local consistencies during feature propagation, bringing in more valuable information while mitigating information loss, as demonstrated in Tables 3 and 4. Further, as shown in Figs. 11 and 12, the cooperation between the AFMA and SPM modules not only overcomes the problem of information loss but also mines and integrates global dependencies and local consistencies, thus obtaining better

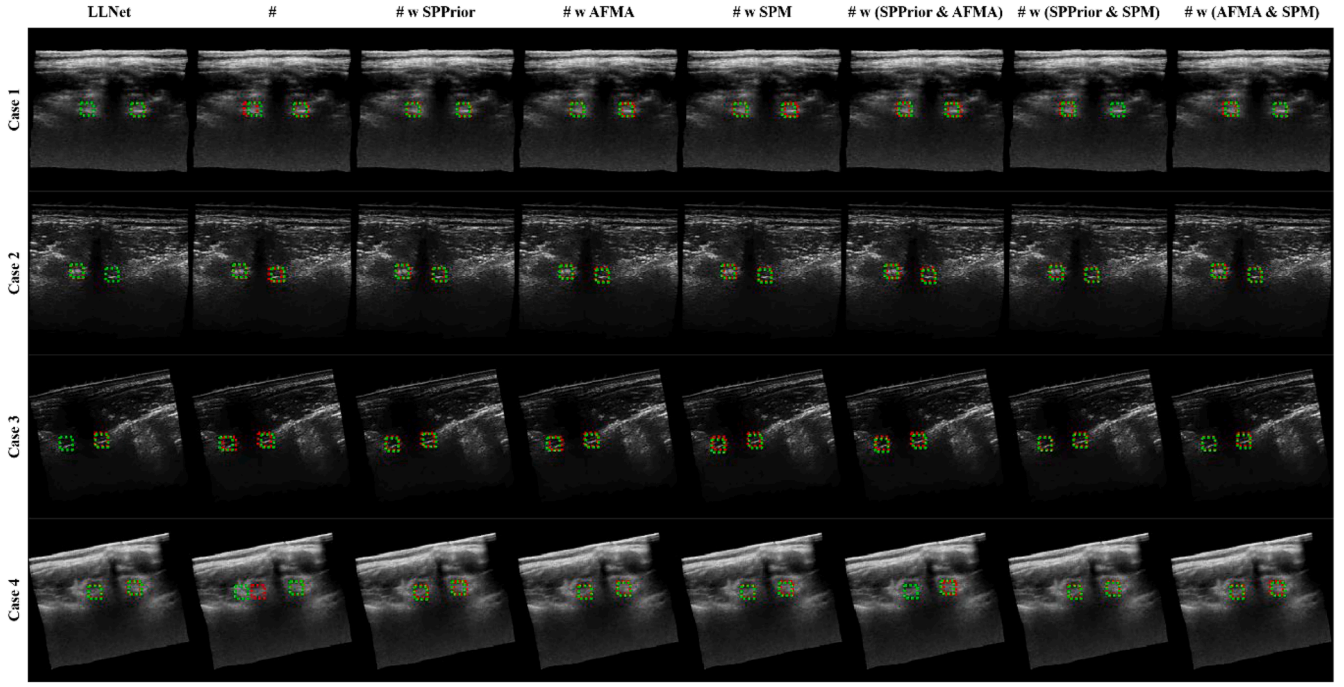


Fig. 12. Visualization examples of lamina detection in ablation studies. Green and red dashed bounding boxes represent the detected ROI containing laminae and corresponding ground truths, representatively. Backbone is symbolized as # and * represents the backbone cooperates with * module.

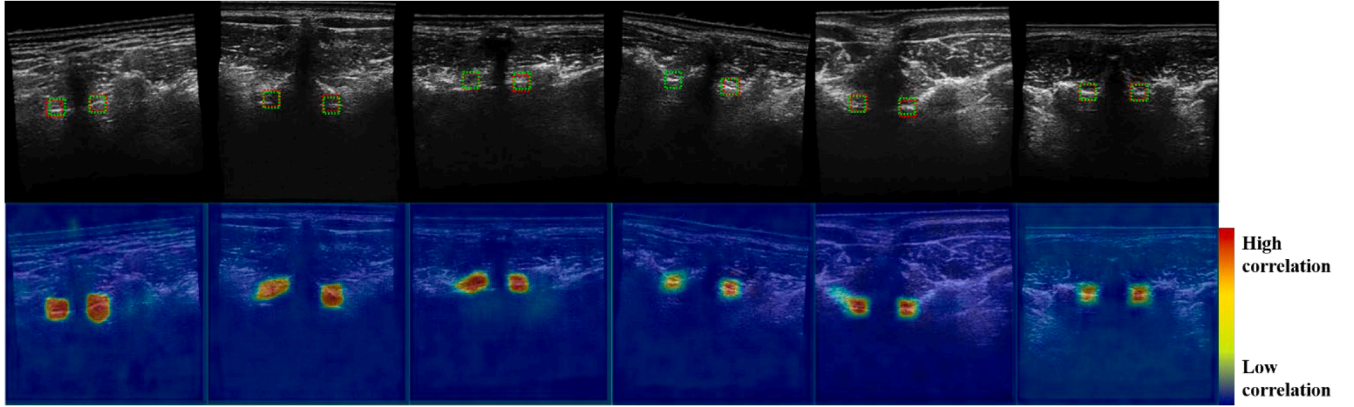


Fig. 13. Visualization examples of refined attention map used for polishing final output. Green and red dashed bounding boxes represent the detected ROI containing laminae and corresponding ground truths, representatively.

results.

6. Discussion

3D spinal ultrasound imaging provides a more flexible option for early screening and longitudinal follow-up without radiation hazards. However, the observability of lamina is likely to be limited due to ultrasonic wave reflection and some interference, such as poor probe contact. A promising solution is to introduce the vertebral anatomical prior into the lamina detector, which suggests the potential location of the lamina through a more observable anatomical component. This is our strategy, too. Our approach is more targeted than those frameworks that enhance the multi-scale feature representation capability or enrich diverse receptive fields. Unlike the lamina, which is located at the valley of the “W-” shape contour as displayed in Fig. 1, the SP is always at the peak, so it is more observable. Further, their anatomical correlation is stable since the vertebra shape is relatively fixed. If the SP can be pre-located and the anatomical correlation can be embedded into the

model, the lamina detection will become more precise. Inspired by these, the LLNet establishes the SP-Prior module to implicitly embed the anatomical correlation between the SP and the lamina and then utilize the correlation to localize the lamina with the assistance of SP prior. As demonstrated in ablation studies, the SP-Prior is a valuable helper in improving lamina detection accuracy. Additionally, the pre-localization of SP is fully automatically achieved in our pipeline, so no additional labor costs are required. The entire framework has the potential to be part of a computer-aided diagnostic system based on 3D spinal ultrasound imaging. We also have other targeted designs for the LLNet to address the challenge of small target detection. The information loss caused by continuous convolutional and pooling operations should be mitigated to ensure a meaningful feature representation of lamina for final decoding. DenseNet has indicated a practical approach for this problem by directly transferring the feature map to the final output layer (Huang et al., 2017). The AFMA module employs a similar solution, and the transferred feature map is further refined by communicating with those features extracted from the original image. Compared with

baseline performance, incorporating the backbone and the AFMA performs better. Global dependencies and contextual information are also critical for obtaining a constructive image pattern, which is beneficial for the model to get a global perception of the whole image. In our framework, the SPM, working on multi-scale features delivered by skip connections, effectively extracts global dependencies and contextual information through updating and fusing the learnable prior via the global attention mechanism. Moreover, evaluation metrics in Table 4 highlight the effectiveness of SPM, too. The three innovations work together to establish an effective lamina localization model, and a more comprehensive description of vertebral rotation and 3D spinal profile can be further achieved based on detected laminae.

Of course, there are still some potential approaches deserve to explore for further enhancing the accuracy of lamina detection. For example, a structural continuity exists among adjacent transverse slices, if we can take advantage of this continuity, like using the detected result of high-image-quality slice to guide that of low-image-quality one, the overall lamina localization will be better and more spatially rational. Additionally, the detection efficiency of our model should be further improved and matched with the imaging speed, for an availability in clinical practices.

7. Conclusion

In this paper, a new detector is proposed to extract anatomical landmarks in 3D spinal ultrasound volume data for a comprehensive assessment of spinal deformity. We innovatively establish a parallel prior embedding branch to explore the anatomical correlation between the lamina and the SP, and then utilize the correlation to optimize the rough detection result. Moreover, the cooperation with two effective attention mechanisms further enhance the capability of our model in small target detection. A comprehensive validation based on data from clinical practices also demonstrates the outstanding capability of our model. In summary, the LLNet has the potential to play a helpful role in a computer-assisted-diagnosis system based on 3D spinal ultrasound imaging.

CRedit authorship contribution statement

Yi Huang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology. **Jing Jiao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology. **Jinhua Yu:** Supervision, Resources, Project administration, Conceptualization. **Yongping Zheng:** Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Yuanyuan Wang:** Supervision, Resources, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that there is not financial interests and personal relationships which may affect the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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